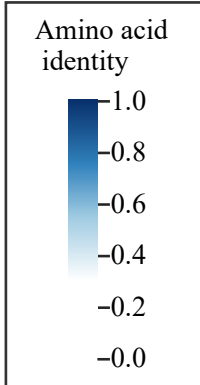


***Sodalinema* sp. AB48**
Group II *nif* cluster

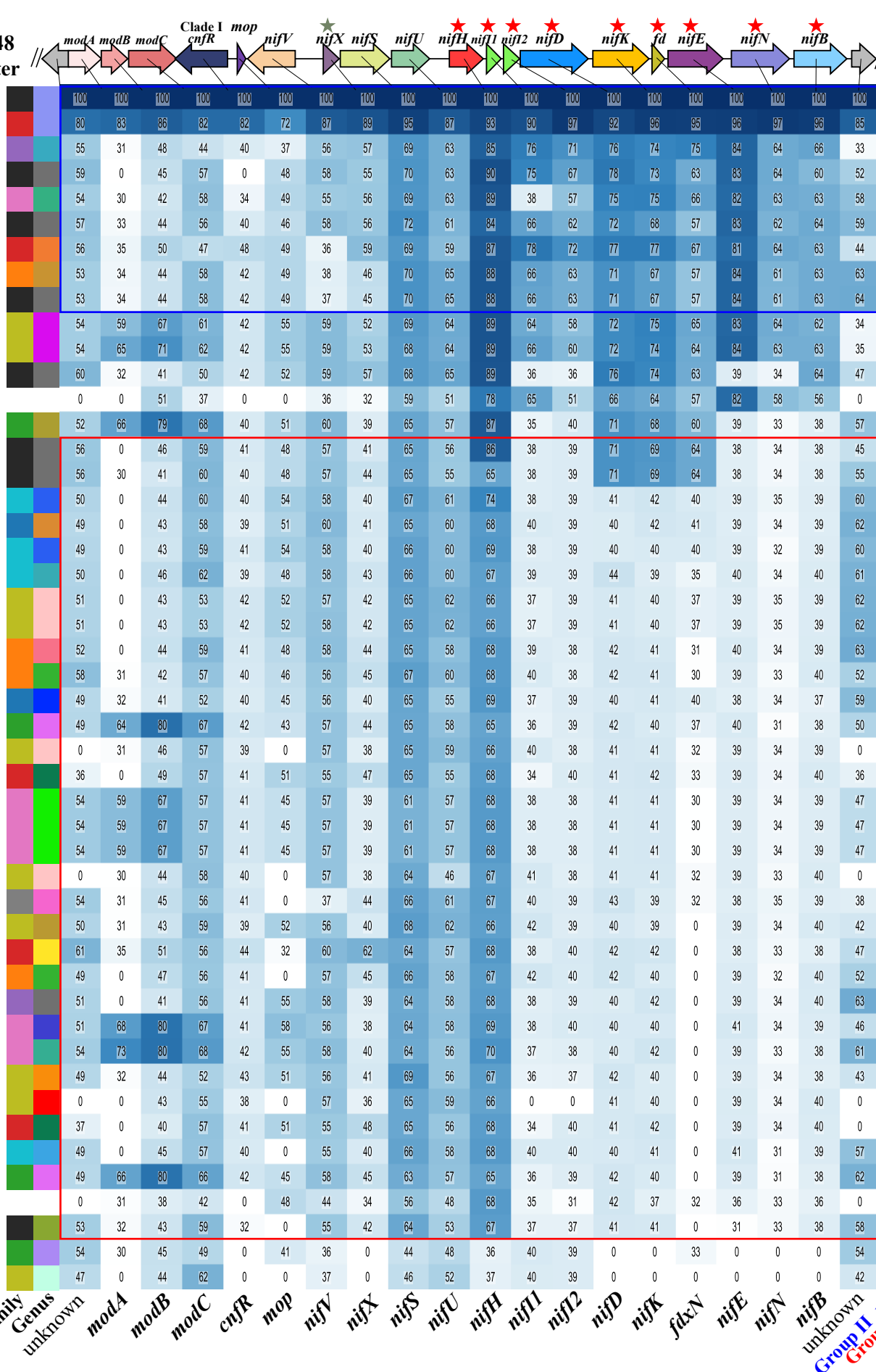


Family

- Chroococcidiopsidaceae
- Coleofasciculaceae
- Elainellaceae
- Geitlerinemataceae
- Geitlerinemataceae_A
- Gloeomargaritaceae
- Leptolyngbyaceae
- Microcoleaceae
- Microcystaceae
- Nostocaceae
- Pseudanabaenaceae
- Rubidibacteraceae
- Synechococcaceae
- Thermostichaceae
- Thermosynechococcaceae
- Unknown
- Xenococcaceae

Genus

- Allocoleopsis*
- Atelocyanobacterium*
- Baaleninema*
- Chroococcidiopsis*
- Coleofasciculus*
- Crocospaera*
- DVEB01
- Geitlerinema*
- Gloeomargarita*
- Gloeotheca*
- Halothece*
- Hydrococcus*
- LEGE-06147
- Leptodesmis*
- Leptolyngbya*
- Leptothermofonsia*
- Nostoc*
- PCC-9228
- Pseudanabaena*
- Raphidiopsis*
- Rippkaea*
- Sodalinema*
- Synechococcus*
- Synechocystis*
- Thermoleptolyngbya*
- Thermostichus*
- Trichodesmium*
- Trichormus*
- Unknown
- Non-cyanobacterial strain



Group II vs Group I

★ enriched in Group II (Bonferroni H <0.01)

★ Forms a Group II-specific clade in ML-tree but is not grouped into a single MCL protein cluster in the PIRATE orthologue classification.

- Sodalinema* sp. AB48 ●
- Phormidium* sp. PBR-2020 ●
- Geitlerinema* sp. PCC 9228 ▲
- Roseofilum* sp. Belize Diploria ▲
- Pleurocapsales cyanobacterium* LEGE 06147 ▲
- Phormidium* sp. CCY1219 ▲
- Baaleninema* sp. ▲
- Coleofasciculus chthonoplastes* PCC 7420 ▲
- Coleofasciculus* sp. A1-SPW-01 ▲
- ▲● *Pleurocapsa* sp. PCC 7327 ●
- *Hydrococcus rivularis* NIES-593 ●
- ▲▲ *Lyngbya* sp. CCY1209 ●
- *Desulfomonile tiedjei* DSM 6799 ●
- ▲● *Synechococcales cyanobacterium* M55 K2018 004 ●
- *Microseira* sp. BLCC-F43 ●
- *Microseira wollei* NIES-4236 ●
- *Trichormus variabilis* ATCC 29413 ●
- *Chroococcidiopsis thermalis* PCC 7203 ●
- *Nostoc* sp. PCC 7120 ●
- *Nostoc* sp. *Peltigera membranacea* cyanobiont N6 ●
- *Rippkaea orientalis* PCC 8801 ●
- *Rippkaea orientalis* PCC 8802 ●
- *Allocoleopsis franciscana* PCC 7113 ●
- *Halothece* sp. PCC 7418 ●
- *Pseudanabaena* sp. ABRG5-3 ●
- *Leptolyngbya* sp. O-77 ●
- Endosymbiont of *Rhopalodia gibberula* ●
- *Synechococcus* sp. JA-2-3B'a(2-13) ●
- *Leptolyngbya boryana* dg5 ●
- *Leptolyngbya boryana* IU 594 ●
- *Leptolyngbya boryana* NIES-2135 ●
- Endosymbiont of *epithemia turgida* ETSB lake ●
- *Trichodesmium erythraeum* IMS101 ●
- *Crocospaera subtropica* ATCC 51142 ●
- *Cyanobacteria bacterium* SBC ●
- *Euhalothece natronophila* Z-M001 ●
- *Cyanothece* sp. PCC 7425 ●
- *Leptothermofonsia sichuanensis* PKUAC-SCTE412 ●
- *Leptodesmis sichuanensis* PKUAC-SCTA121
- *Gloeotheca citrififormis* PCC 7424 ●
- Endosymbiont of *Braarudosphaera bigelowii* ●
- *Synechococcus* sp. JA-3-3Ab
- *Cylindrospermopsis raciborskii* N8 ●
- *Thermoleptolyngbya sichuanensis* PKUAC-SCTA183 ●
- *Azotobacter vinelandii* CA ●
- *Gloeomargaritales cyanobacterium* VI4D9
- Synechococcus elongatus* PCC 7942 ●
- Synechocystis* sp. PCC 6803 ●

▲ draft genome
● complete genome